

Defining causal effects

STSCI / INFO / ILRST 3900: Causal Inference

27 Aug 2026

Learning goals for today

By the end of class, you will be able to

- ▶ explain the fundamental problem of causal inference and the need for causal arguments
- ▶ define potential outcomes

Causal claims hinge on assumptions, not just data



¹Photo credit: YantImages via Wikipedia

Causal claims hinge on arguments, not just data

1. Statistical evidence

- ▶ Alysia Liu wore two skates in Milan. She won a gold medal.

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2. Possible causal claim

- ▶ Wearing two skates in Milan causes a person to win a gold medal.

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2. Possible causal claim

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What do we mean when we say “cause”?

Causal claims hinge on arguments, not just data

	Do you win gold if you:		Causal effect of wearing skates
	Wear skates	Do not wear skates	
Alysa Liu	Yes (1)	?	?
Sam	?	No (0)	?

Causal claims hinge on arguments, not just data

	Do you win gold if you:		Causal effect of wearing skates
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Alysa Liu	Yes (1)	No (0)	?
Sam	?	No (0)	?

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	Do you win gold if you:		Causal effect of wearing skates
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Alysa Liu	Yes (1)	No (0)	+1
Sam	?	No (0)	?

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Alysa Liu	Yes (1)	No (0)	+1
Sam	No (0)	No (0)	0

Observing



Image source: Wikimedia

Observation:

People who eat a Mediterranean diet have lower rates of cardiovascular disease

Observing



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People who eat a Mediterranean diet have lower rates of cardiovascular disease

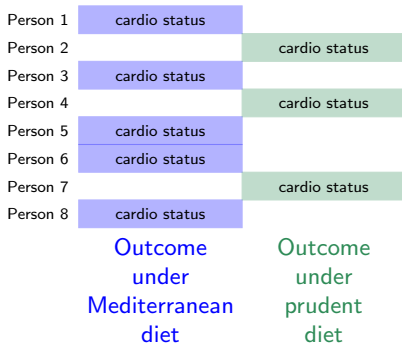
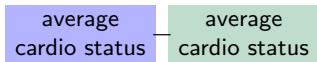
Discuss:

Do you think there is a causal effect here? If not, how might you explain away this apparent observation?

Fundamental problem of causal inference

Holland 1986

Descriptive evidence



Fundamental problem of causal inference

Holland 1986

Descriptive evidence



Causal claim



Person 1	cardio status	
Person 2		cardio status
Person 3	cardio status	
Person 4		cardio status
Person 5	cardio status	
Person 6	cardio status	
Person 7		cardio status
Person 8	cardio status	

Outcome
under
Mediterranean
diet

Outcome
under
prudent
diet

cardio status	cardio status
cardio status	cardio status
cardio status	cardio status
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Fundamental problem of causal inference

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Descriptive evidence



Person 1	cardio status	missing
Person 2	missing	cardio status
Person 3	cardio status	missing
Person 4	missing	cardio status
Person 5	cardio status	missing
Person 6	cardio status	missing
Person 7	missing	cardio status
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Causal claim



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Causal claim



Causal inference is a **missing data** problem

Person 1	cardio status	missing
Person 2	missing	cardio status
Person 3	cardio status	missing
Person 4	missing	cardio status
Person 5	cardio status	missing
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Mathematical notation: Potential outcomes²

²Capital letters and lowercase letters mean different things!

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Y_i (observed/realized) Outcome Whether person i has cardio disease

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Mathematical notation: Potential outcomes²

Y_i	(observed/realized) Outcome	Whether person i has cardio disease
A_i	Treatment	Whether person i ate a Mediterranean diet

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Mathematical notation: Potential outcomes²

Y_i	(observed/realized) Outcome	Whether person i has cardio disease
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Y_i^a	Potential Outcome	Outcome person i would realize if assigned to treatment value a

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Examples:

If assigned prudent diet	If assigned mediterranean diet
Disease	No Disease

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$A_{\text{Sam}} = \text{MedDiet}$ We observed that Sam ate a Mediterranean diet

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 $Y_{\text{Sam}} = \text{survived}$ We observe that Sam has no disease

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$Y_{\text{Sam}} = \text{survived}$ We observe that Sam has no disease

$Y_{\text{Sam}}^{\text{MedDiet}} = \text{survived}$ If Sam had been assigned a Mediterranean diet
he would have no disease

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$Y_{\text{Sam}} = \text{survived}$ We observe that Sam has no disease

$Y_{\text{Sam}}^{\text{MedDiet}} = \text{survived}$ If Sam had been assigned a Mediterranean diet
he would have no disease

$Y_{\text{Sam}}^{\text{PruDiet}} = \text{died}$ If Sam had been assigned a prudent diet
he would have disease

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Practice

Using the slip of paper you received and the diet you follow, what is

- ▶ Y_i
- ▶ A_i
- ▶ Y_i^{MedDiet}
- ▶ Y_i^{PruDiet}

The consistency assumption

- **Consistency Assumption:** the observed outcome for an individual corresponds to the potential outcome for that individual's observed treatment

$$\underbrace{Y_i}_{\text{Observed Outcome}} = \underbrace{Y_i^{A_i}}_{\text{Potential Outcome for observed treatment}}$$

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- Consistency implies that observed outcome for an individual who “chose” a specific treatment is same as what we would have observed if the individual was “assigned” that treatment
- Consistency would not hold if “forcing” someone to eat a Mediterranean diet results in a different lifespan than what we observed for someone who chose the Mediterranean diet

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- Consistency would not hold if “forcing” someone to eat a Mediterranean diet results in a different lifespan than what we observed for someone who chose the Mediterranean diet
- Consistency implies treatments don't have hidden types
- Consistency would not hold if Italian olive oil resulted in a different lifespan than Greek olive oil

Practice: How would you say this in English?

We might wonder how a person's earnings relate to whether they hold a college degree

1. $E(\text{Earnings} \mid \text{Degree} = \text{TRUE}) > E(\text{Earnings} \mid \text{Degree} = \text{FALSE})$

2. $E(\text{Earnings}^{\text{Degree}=\text{TRUE}}) > E(\text{Earnings}^{\text{Degree}=\text{FALSE}})$

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► Average earnings are higher among those with college degrees

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- ▶ Average earnings are higher among those with college degrees

2. $E(\text{Earnings}^{\text{Degree}=\text{TRUE}}) > E(\text{Earnings}^{\text{Degree}=\text{FALSE}})$

- ▶ The average earning if everyone was assigned to get a degree is higher than the average earnings if everyone was assigned to not get a degree
- ▶ On average, getting a degree causes higher earnings

Practice:

1. On average, students who do the homework learn more than those who don't
2. On average, doing the homework causes more learning

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1. On average, students who do the homework learn more than those who don't

$$E(\text{Learning} \mid \text{HW} = \text{TRUE}) > E(\text{Learning} \mid \text{HW} = \text{FALSE})$$

2. On average, doing the homework causes more learning

$$E(\text{Learning}^{\text{HW}=\text{TRUE}}) > E(\text{Learning}^{\text{HW}=\text{FALSE}})$$

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You can now

- ▶ Read Chapter 1 of [Hernán and Robins 2020](#)